

G3 Applications of differentiation

Name: _____

Class: _____

Date: _____

Time: **383 minutes**

Marks: **331 marks**

Comments:

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Q1.

State the interval for which $\sin x$ is a decreasing function for $0^\circ \leq x \leq 360^\circ$

(Total 2 marks)

Q2.

A curve has equation $y = 2x^2 - 8x\sqrt{x} + 8x + 1$ for $x \geq 0$

(a) Prove that the curve has a maximum point at $(1, 3)$

Fully justify your answer.

(9)

(b) Find the coordinates of the other stationary point of the curve and state its nature.

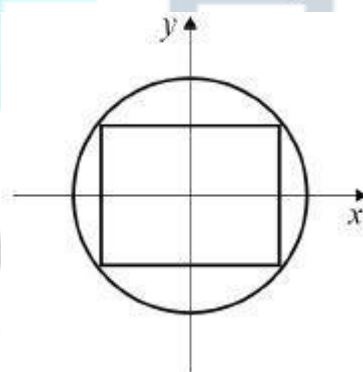
(2)

(Total 11 marks)

Q3.

A company is designing a logo. The logo is a circle of radius 4 inches with an inscribed rectangle. The rectangle must be as large as possible.

The company models the logo on an x - y plane as shown in the diagram.



Use calculus to find the maximum area of the rectangle.

Fully justify your answer.

(Total 10 marks)

Q4.

A curve has equation $y = x^3 - 48x$

The point A on the curve has x coordinate -4

The point B on the curve has x coordinate $-4 + h$

(a) Show that the gradient of the line AB is $h^2 - 12h$

(4)

- (b) Explain how the result of part (a) can be used to show that A is a stationary point on the curve.

(2)

(Total 6 marks)

Q5.

A curve has equation $y = x^5 + 4x^3 + 7x + q$ where q is a positive constant.

Find the gradient of the curve at the point where $x = 0$

Circle your answer.

0 4 7 q

(Total 1 mark)

Q6.

A function f is defined by $f(x) = \frac{x}{\sqrt{2x-2}}$

- (a) State the maximum possible domain of f .

(2)

- (b) Use the quotient rule to show that $f'(x) = \frac{x-2}{(2x-2)^{\frac{3}{2}}}$

(3)

- (c) Show that the graph of $y = f(x)$ has exactly one point of inflection.

(7)

- (d) Write down the values of x for which the graph of $y = f(x)$ is convex.

(1)

(Total 13 marks)

Q7.

Chris claims that, "for any given value of x , the gradient of the curve

$y = 2x^3 + 6x^2 - 12x + 3$ is always greater than the gradient of the curve $y = 1 + 60x - 6x^2$."

Show that Chris is wrong by finding all the values of x for which his claim is **not** true.

(Total 7 marks)

Q8.

A curve has equation $y = 6x\sqrt{x} + \frac{32}{x}$ for $x > 0$

(a) Find $\frac{dy}{dx}$

(4)

(b) The point A lies on the curve and has x -coordinate 4

Find the coordinates of the point where the tangent to the curve at A crosses the x -axis.

(5)

(Total 9 marks)

Q9.

Find the coordinates, in terms of a , of the minimum point on the curve $y = x^2 - 5x + a$, where a is a constant.

Fully justify your answer.

(Total 3 marks)

Q10.

Prove that the function $f(x) = x^3 - 3x^2 + 15x - 1$ is an increasing function.

(Total 6 marks)

Q11.

A curve has equation $y = \frac{2x+3}{4x^2+7}$

(a) (i) Find $\frac{dy}{dx}$

(2)

(ii) Hence show that y is increasing when $4x^2 + 12x - 7 < 0$

(4)

(b) Find the values of x for which y is increasing.

(2)

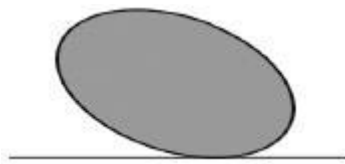
(Total 8 marks)

Q12.

A sculpture formed from a prism is fixed on a horizontal platform, as shown in the diagram.

The shape of the cross-section of the sculpture can be modelled by the equation $x^2 + 2xy + 2y^2 = 10$, where x and y are measured in metres.

The x and y axes are horizontal and vertical respectively.



Find the maximum vertical height above the platform of the sculpture.

(Total 8 marks)

Q13.

An open-topped fish tank is to be made for an aquarium.
It will have a square horizontal base, rectangular vertical sides and a volume of 60 m^3
The materials cost:

- £15 per m^2 for the base
 - £8 per m^2 for the sides.
- (a) Modelling the sides and base of the fish tank as laminae, use calculus to find the height of the tank for which the overall cost of the materials has its minimum value.

Fully justify your answer.

(8)

- (b) (i) **In reality, the thickness of the base and sides of the tank is 2.5 cm**

Briefly explain how you would refine your modelling to take account of the thickness of the sides and base of the tank.

(1)

- (ii) How would your refinement affect your answer to part (a)?

(1)

(Total 10 marks)

Q14.

A curve is defined by the parametric equations

$$x = t^3 + 2, \quad y = t^2 - 1$$

- (a) Find the gradient of the curve at the point where $t = -2$
- (b) Find a Cartesian equation of the curve.

(4)

(2)

(Total 6 marks)

Q15.

A curve has equation $y = 2x \cos 3x + (3x^2 - 4) \sin 3x$

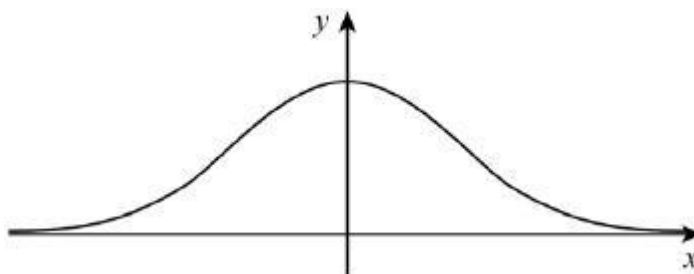
- (a) Find $\frac{dy}{dx}$, giving your answer in the form $(mx^2 + n) \cos 3x$, where m and n are integers. (4)
- (b) Show that the x -coordinates of the points of inflection of the curve satisfy the equation

$$\cot 3x = \frac{9x^2 - 10}{6x}$$

(4)
(Total 8 marks)

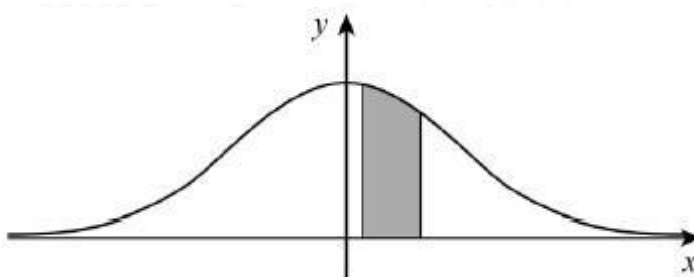
Q16.

The diagram shows part of the graph of $y = e^{-x^2}$



The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

- (a) Find the values of x for which the graph is concave. (4)
- (b) The finite region bounded by the x -axis and the lines $x = 0.1$ and $x = 0.5$ is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2} dx$

Give your estimate to four decimal places.

- (3)
- (c) Explain with reference to your answer in part (a), why the answer you found in part (b) is an underestimate.

(2)

- (d) By considering the area of a rectangle, and using your answer to part (b), prove that the shaded area is 0.4 correct to 1 decimal place.

(3)

(Total 12 marks)

Q17.

A bird flies from a tree. At time t seconds, the bird's height, y metres, above the horizontal ground is given by

$$y = \frac{1}{8}t^4 - t^2 + 5, \quad 0 \leq t \leq 4$$

- (a) Find $\frac{dy}{dt}$. (2)

- (b) (i) Find the rate of change of height of the bird in metres per second when $t = 1$. (2)

- (ii) Determine, with a reason, whether the bird's height above the horizontal ground is increasing or decreasing when $t = 1$. (1)

- (c) (i) Find the value of $\frac{d^2y}{dt^2}$ when $t = 2$. (2)

- (ii) Given that y has a stationary value when $t = 2$, state whether this is a maximum value or a minimum value. (1)

(Total 8 marks)

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Q18.

The gradient, $\frac{dy}{dx}$, of a curve at the point (x, y) is given by

$$\frac{dy}{dx} = 10x^4 - 6x^2 + 5$$

The curve passes through the point $P(1, 4)$.

- (a) Find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$. (3)
- (b) Find the equation of the curve.

(5)
(Total 8 marks)

Q19.

The point $P(2, 8)$ lies on a curve, and the point M is the only stationary point of the curve.

The curve has equation $y = 6 + 2x - \frac{8}{x^2}$.

- (a) Find $\frac{dy}{dx}$. (3)
- (b) Show that the normal to the curve at the point $P(2, 8)$ has equation $x + 4y = 34$. (3)
- (c) (i) Show that the stationary point M lies on the x -axis. (3)
- (ii) Hence **write down** the equation of the tangent to the curve at M . (1)
- (d) The tangent to the curve at M and the normal to the curve at P intersect at the point T . Find the coordinates of T . (2)

(Total 12 marks)

Q20.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 - 4x + 15$.
- (i) Use the Factor Theorem to show that $x + 3$ is a factor of $f(x)$. (2)
- (ii) Express $f(x)$ in the form $(x + 3)(x^2 + px + q)$, where p and q are integers. (2)
- (b) A curve has equation $y = x^4 - 8x^2 + 60x + 7$.
- (i) Find $\frac{dy}{dx}$. (3)
- (ii) Show that the x -coordinates of any stationary points of the curve satisfy the equation

$$x^3 - 4x + 15 = 0$$

(1)

- (iii) Use the results above to show that the only stationary point of the curve occurs when $x = -3$. (2)
- (iv) Find the value of $\frac{d^2y}{dx^2}$ when $x = -3$. (3)
- (v) Hence determine, with a reason, whether the curve has a maximum point or a minimum point when $x = -3$. (1)
- (Total 14 marks)

Q21.

A curve has the equation

$$y = \frac{12 + x^2\sqrt{x}}{x}, \quad x > 0$$

- (a) Express $\frac{12 + x^2\sqrt{x}}{x}$ in the form $12x^p + x^q$. (3)
- (b) (i) Hence find $\frac{dy}{dx}$. (2)
- (ii) Find an equation of the normal to the curve at the point on the curve where $x = 4$. (4)
- (iii) The curve has a stationary point P . Show that the x -coordinate of P can be written in the form 2^k , where k is a rational number. (3)
- (Total 12 marks)

Q22.

- (a) Given that $y = x^4 \tan 2x$, find $\frac{dy}{dx}$. (3)
- (b) Find the gradient of the curve with equation $y = \frac{x^2}{x-1}$ at the point where $x = 3$. (3)
- (Total 6 marks)

Q23.

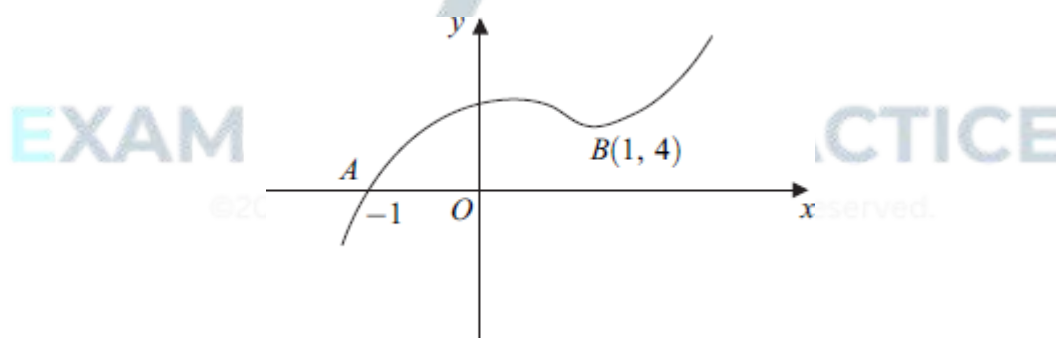
A curve is defined by the parametric equations $x = 8e^{-2t} - 4$, $y = 2e^{2t} + 4$.

- (a) Find $\frac{dy}{dx}$ in terms of t . (3)
- (b) The point P , where $t = \ln 2$, lies on the curve.
- (i) Find the gradient of the curve at P . (1)
- (ii) Find the coordinates of P . (2)
- (iii) The normal at P crosses the x -axis at the point Q . Find the coordinates of Q . (3)
- (c) Find the Cartesian equation of the curve in the form $xy + 4y - 4x = k$, where k is an integer. (3)

(Total 12 marks)

Q24.

The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



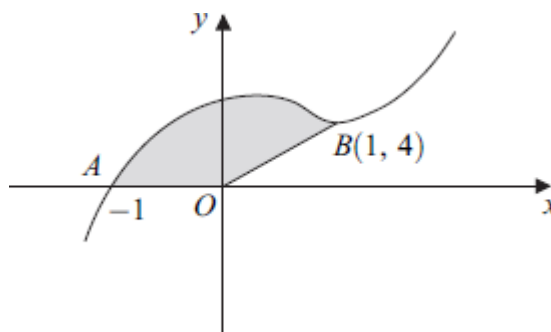
- (a) Given that $y = x^5 - 3x^2 + x + 5$, find:
- (i) $\frac{dy}{dx}$; (3)
- (ii) $\frac{d^2y}{dx^2}$. (1)
- (b) Find an equation of the tangent to the curve at the point $A(-1, 0)$.

(2)

- (c) Verify that the point B , where $x = 1$, is a minimum point of the curve.

(3)

- (d) The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



- (i) Find $\int_{-1}^1 (x^5 - 3x^2 + x + 5) dx$.

(5)

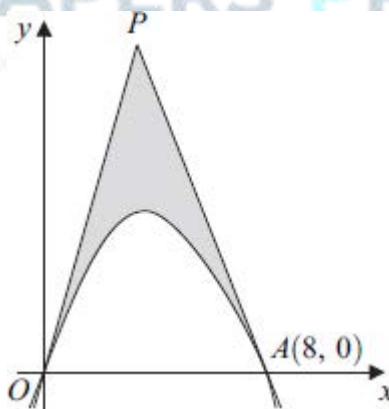
- (ii) Hence find the area of the shaded region bounded by the curve between A and B and the line segments AO and OB .

(2)

(Total 16 marks)

Q25.

The diagram shows part of a curve crossing the x -axis at the origin O and at the point $A(8, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve has equation

$$y = 12x - 3x^{\frac{5}{3}}$$

- (a) Find $\frac{dy}{dx}$. (2)
- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)
- (ii) Show that the equation of the tangent at $A(8, 0)$ is $y + 8x = 64$. (3)
- (c) Find $\int (12x - 3x^{\frac{5}{3}}) dx$. (3)
- (d) Calculate the area of the shaded region bounded by the curve from O to A and the tangents OP and AP . (7)
- (Total 17 marks)**

Q26.

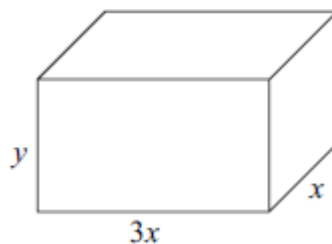
A curve is defined by the parametric equations

$$x = 8t^2 - t, \quad y = \frac{3}{t}$$

- (a) Show that the cartesian equation of the curve can be written as $xy^2 + 3y = k$, stating the value of the integer k . (2)
- (b) (i) Find an equation of the tangent to the curve at the point P , where $t = \frac{1}{4}$. (7)
- (ii) Verify that the tangent at P intersects the curve when $x = \frac{3}{2}$. (2)
- (Total 11 marks)**

Q27.

The diagram shows a solid cuboid with sides of lengths x cm, $3x$ cm and y cm.



The total surface area of the cuboid is 32 cm^2 .

- (a) (i) Show that $3x^2 + 4xy = 16$.

(2)

- (ii) Hence show that the volume, $V \text{ cm}^3$, of the cuboid is given by

$$V = 12x - \frac{9x^3}{4}$$

(2)

- (b) Find $\frac{dV}{dx}$.

(2)

- (c) (i) Verify that a stationary value of V occurs when $x = \frac{4}{3}$.

(2)

- (ii) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = \frac{4}{3}$.

(2)

(Total 10 marks)

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Q28.

The gradient, $\frac{dy}{dx}$, of a curve C at the point (x, y) is given by

$$\frac{dy}{dx} = 20x - 6x^2 - 16$$

- (a) (i) Show that y is increasing when $3x^2 - 10x + 8 < 0$.

(2)

- (ii) Solve the inequality $3x^2 - 10x + 8 < 0$.

(4)

- (b) The curve C passes through the point $P(2, 3)$.

- (i) Verify that the tangent to the curve at P is parallel to the x -axis.

(2)

- (ii) The point $Q(3, -1)$ also lies on the curve. The normal to the curve at Q and the tangent to the curve at P intersect at the point R . Find the coordinates of R .

(7)

(Total 15 marks)

Q29.

At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^2 - \frac{4}{x^2} - 11$$

The point $P(2, 1)$ lies on the curve.

- (a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 2$.

(1)

- (ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 2$.

(4)

- (iii) Hence state whether P is a maximum point or a minimum point, giving a reason for your answer.

(1)

- (b) Find the equation of the curve.

(4)

(Total 10 marks)

Q30.

A curve has equation $y = x^3 \ln x$.

- (a) Find $\frac{dy}{dx}$.

(2)

- (b) (i) Find an equation of the tangent to the curve $y = x^3 \ln x$ at the point on the curve where $x = e$.

(3)

- (ii) This tangent intersects the x -axis at the point A . Find the exact value of the x -coordinate of the point A .

(2)

(Total 7 marks)

Q31.

- (a) Given that $x = \frac{\sin y}{\cos y}$, use the quotient rule to show that

$$\frac{dx}{dy} = \sec^2 y \quad (3)$$

- (b) Given that $\tan y = x - 1$, use a trigonometrical identity to show that

$$\sec^2 y = x^2 - 2x + 2 \quad (2)$$

- (c) Show that, if $y = \tan^{-1}(x - 1)$, then

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2} \quad (1)$$

- (d) A curve has equation $y = \tan^{-1}(x - 1) - \ln x$.

- (i) Find the value of the x -coordinate of each of the stationary points of the curve. (4)

- (ii) Find $\frac{d^2y}{dx^2}$. (2)

- (iii) Hence show that the curve has a minimum point which lies on the x -axis. (2)

(Total 14 marks)

Q32.

A curve is defined by the parametric equations

$$x = 2 \cos \theta, \quad y = 3 \sin 2\theta$$

- (a) (i) Show that

$$\frac{dy}{dx} = a \sin \theta + b \operatorname{cosec} \theta$$

where a and b are integers. (4)

- (ii) Find the gradient of the normal to the curve at the point where $\theta = \frac{\pi}{6}$. (2)

- (b) Show that the cartesian equation of the curve can be expressed as

$$y^2 = px^2(4 - x^2)$$

where p is a rational number.

(3)

(Total 9 marks)

Q33.

A curve is defined by the equation $9x^2 - 6xy + 4y^2 = 3$.

Find the coordinates of the two stationary points of this curve.

(Total 8 marks)

Q34.

The curve with equation $y = 13 + 18x + 3x^2 - 4x^3$ passes through the point P where $x = -1$.

- (a) Find $\frac{dy}{dx}$

(3)

- (b) Show that the point P is a stationary point of the curve and find the other value of x where the curve has a stationary point

(3)

- (c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P .

(3)

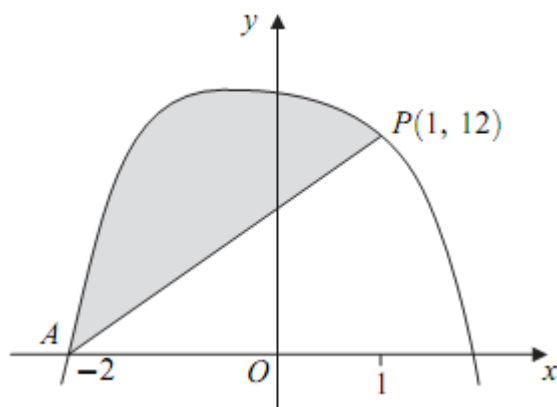
- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)

(Total 10 marks)

Q35.

The curve sketched below passes through the point $A (-2, 0)$.



The curve has equation $y = 14 - x - x^4$ and the point $P(1, 12)$ lies on the curve.

- (a) (i) Find the gradient of the curve at the point P . (3)
- (ii) Hence find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$. (2)
- (b) (i) Find $\int_{-2}^1 (14 - x - x^4) dx$. (5)
- (ii) Hence find the area of the shaded region bounded by the curve $y = 14 - x - x^4$ and the line AP . (2)
- (Total 12 marks)**